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Middle School STEM
Mary, Star of the Sea School
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MSOS 2026-2027

STEM — Grade 8 Syllabus

Grade 8: Physical Science • SY 2026-2027

Mary, Star of the Sea School Mission

Nurture, Unite, Inspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

2026-2027 Essential Question

"How can I model kindness, fairness, and humility like Jesus?"

Course Description

8th Grade STEM is an introduction to Chemistry and Physics. The course is focused on ways to unite our world view as citizens of the planet, nurture our understanding of how chemistry and physics are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short introduction to the Scientific Method and the Engineering Design Process as well as the principal tools and techniques of a scientist and engineer. Then it's time to launch into the study of matter, the elements, the principles of chemistry, and finally an investigation of polymers and the impact they are having on our environment. After the Winter Break, our third unit will find us going back in time, to discover how the science of Physics developed. We will investigate the laws of motion developed by Sir Isaac Newton and build a few simple machines to throw things around the lab. After Spring Break, we will conclude the year with an investigation of Faraday's Fundamental Forces, electricity and magnetism, and how these two fundamental forces act together.

Teacher

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Classroom & Lab Materials

- Composition notebook (lab journal)
- Pencils, pens, colored pencils
- Lab safety goggles
- Computer / device for simulations
- Calculator

Scope & Sequence — Grade 8: Physical Science

2026–2027, August–June. All unit dates are projected and may be subject to change.

Unit 1 — I am a Scientist and an Engineer 3 Aug - 28 Aug

Key Concept: Systems · Related: Patterns, Form, Function · Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Know: Scientific Method, Engineering Design Process, Properties of Matter.

Understand: How the Scientific Method/Engineering Design Process solves problems. Review the linkage between Mass, Volume, and Density. Investigate heat energy.

Do: Lab Safety, Scientific Method/Experimental Cycle, variables and hypothesis formulation, Lab report format, intro to probe-ware.

Summative Assessment Tasks:

- Properties of Matter Lab (Crit. B/C) Cross-Listed with Math

Unit 2 — A Period Matters! 31 Aug - 18 Dec

Key Concept: Change · Related: Models, Patterns, Consequences · Global Context: Science and Technical Innovation

Inquiry statement: Scientists use models to understand the patterns of change in matter and the consequences of science and technology on our world.

Factual: How are elements organized in the periodic table?

Conceptual: Why is Conservation of Matter so important in chemistry?

Debatable: Should society have control/approval over scientific research?

Know: Structure and states of matter, elements of the Periodic Table, atomic models, types of chemical reactions, field study methods.

Understand: How the atomic theory of interactions is the basis for chemical reactions. How to translate theory into practical application (stoichiometry).

Do: Metrics Lab, Lab Skills Assessment, Chemical Reactions Lab, Acids/Bases Lab, Nuts and Bolts Lab, Stain Lab, Polymer Lab.

Summative Assessment Tasks:

- Element Babybook (Crit. A/D) Cross-Listed with Math
- Berth Project (Crit. A/D)
- Stain Lab (Crit. B/C)

Unit 3 — Forces and Motion 7 Jan – 12 Mar

Key Concept: Change · Related: Transformation, Movement, Energy · Global Context: Orientation in Space and Time

Inquiry statement: Observing changes in movement allows scientists to detect and understand the transformation of energy.

Factual: Why are “unbalanced” forces so important in the understanding of physics?

Conceptual: How are the Laws of Conservation of Motion and Energy related?

Debatable: Must religion and science always be at odds with one another?

Know: Newton’s Laws of Motion, what is velocity and acceleration, Newton’s Universal Law of Gravity.

Understand: How unbalanced forces result in the changes we observe in physics.

Do: Simple Machines Lab, Mover Lab, Velocity Lab, Acceleration Lab, Projectile Lab.

Summative Assessment Tasks:

- The Apprentices Project (Crit. A/B/C/D) Cross-Listed with Math

STEM Assessment Criteria

- A — Knowing & Understanding. Recall and apply scientific knowledge; explain phenomena. (0–8)
- B — Inquiring & Designing. 1st half of the Experimental / Design Cycle: question, plan, hypothesize, design. (0–8)
- C — Processing & Evaluating. 2nd half of the Experimental / Design Cycle: collect, analyze, draw conclusions. (0–8)

- D — Reflecting on the Impacts of Science / Engineering. Consider ethical, environmental, and social implications of science and engineering decisions. (0-8)

Learning Support & Communication

Parents with questions or concerns may directly contact the instructor.

Students are encouraged to use study hall or lunch recess to obtain additional learning support.

The Innovations Lab is open every Wednesday afternoon (except Mini-days) until 4 pm for any student needing help or to finish lab work.

Students and/or parents should email Mr. David Eachus to make arrangements for additional learning opportunities.

Empowered to Succeed, Guided by Faith